

Hina Mehboob

Full-Stack AI Developer | Software Engineering Undergraduate

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PROFILE

Software Engineering undergraduate (7th semester) with hands-on experience in full-stack web development using Next.js and TypeScript, automated testing, and machine learning. Proven ability to build production-grade applications — from AI-powered autonomous pipelines to responsive UIs — through real-world internship and project work. Actively seeking internship or entry-level opportunities in web development.

TECHNICAL SKILLS

Frontend & Backend: Next.js (App Router), TypeScript, React, REST APIs, Tailwind CSS

AI & Automation: LangChain, LangGraph, n8n Agent Orchestration, Anthropic Claude API, Google Gemini API, Supabase Real-time

AI/RAG: Semantic Vector Search, Reciprocal Rank Fusion (RRF), Agentic RAG, Function Calling

Testing: Selenium, Cypress, Playwright

DevOps & Cloud: Docker, GitHub REST API, Vercel REST API, CI/CD Pipelines, Supabase (Postgres), pgvector

Backend: Python, FastAPI, Uvicorn, PostgreSQL, pgvector, Git API

Design: UI/UX Design (Figma), Responsive Design

ML & Python: Python, Scikit-Learn, Random Forest, SVM, Flask

CS Fundamentals: Data Structures & Algorithms, OOP

PROFESSIONAL EXPERIENCE

AI/ML Intern | Rise Tech

Jun 2026 – Present • Islamabad, Pakistan

- ▶ Developing and deploying machine learning models and AI-powered features as part of the core engineering team.
- ▶ Collaborating on NLP and LLM integration projects, leveraging modern AI frameworks and APIs.
- ▶ Contributing to data pipelines, model evaluation, and production deployment workflows.

Software Engineering Intern | InoTech Solutions

Jul 2025 – Aug 2025 • Rawalpindi, Pakistan

- ▶ Built and shipped features for Next.js web applications used in live company projects, including authentication flows and REST API integrations.
- ▶ Designed and executed automated test suites using Selenium, Playwright, and Cypress; produced a comparative analysis to recommend the optimal framework for production use.
- ▶ Created high-fidelity Figma prototypes that informed UI/UX decisions across multiple client projects.

PROJECTS

RepoGazer — AI-Powered Codebase Search Engine

React · Next.js 16 · TypeScript · Python · FastAPI · Supabase · PostgreSQL (pgvector, FTS) · LangChain/LangGraph · Google Gemini API · TailwindCSS

- ▶ Developed an AI-powered codebase search engine that performs semantic code analysis and answers technical developer queries in real-time using a self-correcting Agentic RAG pipeline.
- ▶ Designed and orchestrated a multi-node codebase reasoning graph using LangGraph to automate context retrieval, relevance grading (Pydantic structured output), query rewriting, and LLM synthesis.
- ▶ Engineered a hybrid search algorithm combining pgvector cosine similarity and PostgreSQL Full-Text Search (FTS), applying Reciprocal Rank Fusion (RRF) via PL/pgSQL stored procedures to double search accuracy.
- ▶ Created an AST-aware repository indexing engine using GitPython (shallow cloning) and RecursiveCharacterTextSplitter to dynamically chunk and embed source files using the Gemini Embedding API.

- ▶ Programmed a premium React interface using Next.js 16 (Turbopack), featuring an interactive file-tree explorer, real-time citation highlights, and a syntax-highlighted code workspace.
- ▶ Implemented Server-Sent Events (SSE) in FastAPI to stream LLM responses and source citations in real-time, reducing time-to-first-token (TTFT) by over 70%.

Autonomous AI Web Developer & DevOps Pipeline

Next.js · TypeScript · n8n · LangChain · Claude 3.5 Sonnet · Supabase · GitHub/Vercel APIs · Docker · Tailwind CSS

- ▶ Engineered an event-driven, autonomous AI agent orchestration pipeline using n8n and Anthropic Claude 3.5 Sonnet that fully generates, commits, and provisions complete full-stack web applications from raw natural language prompts.
- ▶ Architected a resilient Human-in-the-Loop (HITL) checkpoint system that manages state persistence, pauses for multi-platform credential validation (GitHub/Vercel OAuth), and dynamically resumes complex multi-stage workflows.
- ▶ Built an algorithmic file-tree generator that parses LLM structural outputs into structured JSON, handles Base64 conversion streams, and automates multi-file commits via the GitHub REST API.
- ▶ Optimized a live cloud-provisioning loop using the Vercel REST API and Supabase Real-time Postgres listeners to stream deployment status logs to a Next.js dashboard, reducing UI latency to near-zero.
- ▶ Implemented rigorous exception handling and asynchronous self-correction logic to automatically detect and repair unclosed code structures before production deployment.

FPSC Official Website

Next.js · Figma · CMS

- ▶ Contributed to the frontend development and CMS management of the official FPSC Pakistan website under direct supervisor guidance.
- ▶ Created UI/UX designs in Figma and developed responsive frontend components using Next.js.

Automation Testing Demo Platform

Next.js · Selenium · Cypress · Playwright

- ▶ Developed a Next.js web application with authentication and API integration as a controlled environment for comparative automated testing.
- ▶ Conducted systematic evaluation of Selenium, Cypress, and Playwright to identify the most suitable testing framework for the company's production workflow.

Reading Tracker — AI-Powered Research App

Next.js · TypeScript · Supabase · AI Chatbot

- ▶ Engineered a full-stack reading tracker with real-time data persistence, automated auto-save note-taking, and an AI-powered chatbot for interactive content analysis.
- ▶ Integrated PDF rendering and managed the full software lifecycle via GitHub with a CI/CD pipeline deployed on Vercel.

Live Cricket Score Predictor

Python · Random Forest · Flask

- ▶ Engineered an end-to-end ML pipeline using Random Forest and Flask to predict live T20 cricket scores, achieving 97% accuracy by modeling stochastic scoring patterns.
- ▶ Developed a real-time web interface providing live score projections during matches.

HamOrSpam Email Detector

Python · SVM · Logistic Regression · Flask

- ▶ Implemented SVM and Logistic Regression from scratch to classify SMS spam using multiple loss functions; built a real-time Flask dashboard with live training visualization and inference.
- ▶ Compared manual ML implementations against Scikit-Learn, analyzing optimization behavior and performance trade-offs.

EDUCATION

B.E. Software Engineering | National University of Sciences and Technology (NUST) Oct 2023 – Oct 2027 • Rawalpindi, Pakistan

Relevant coursework: Data Structures & Algorithms, OOP, Database Systems, Software Engineering Principles, Machine Learning